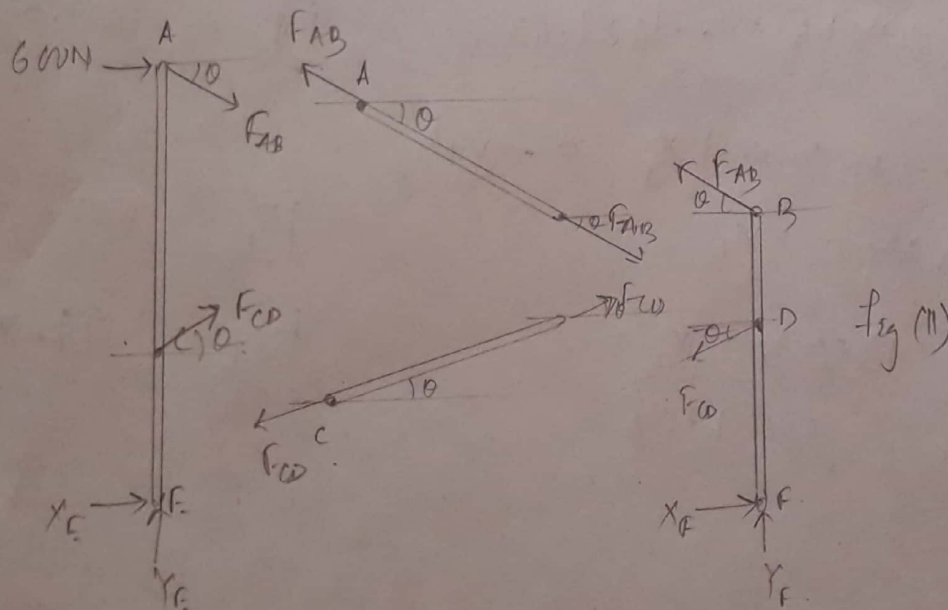
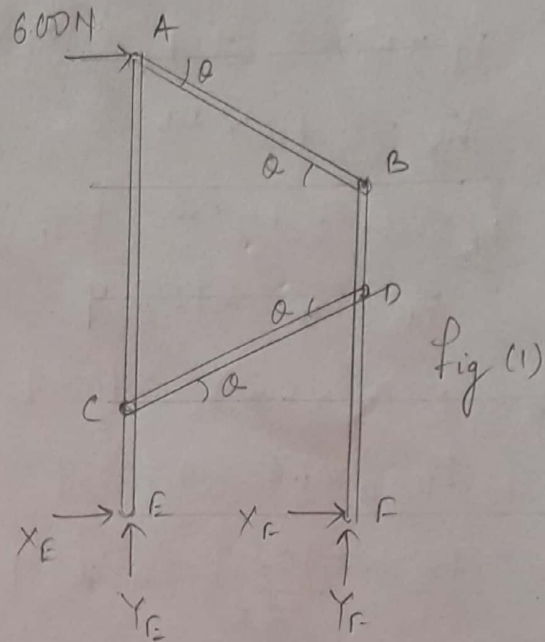
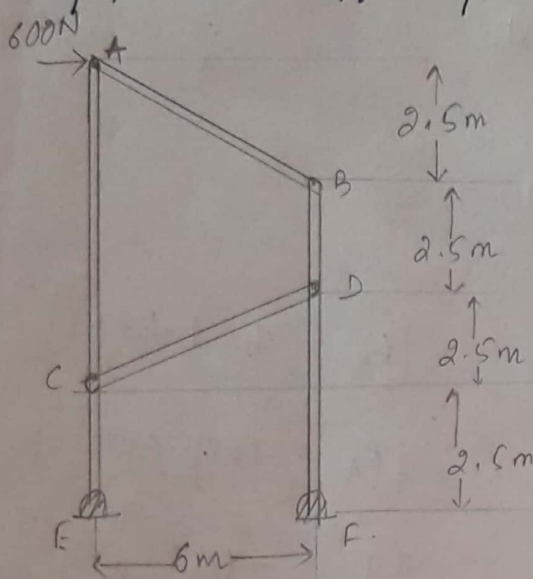


FRAMES

Exercise-1

Calculate the forces in pins A, B, C, D, E and F of the frame loaded and supported as shown.



In Figure. (I)

$$\sum X = X_E + X_F + 600 = 0 \quad \text{--- (I)}$$

$$\sum Y = Y_E + Y_F = 0 \quad \text{--- (II)}$$

$$\sum M_E = -600 \times 10 + Y_F \times 6 = 0$$

$$\boxed{Y_F = 1000 \text{ N}}$$

$$\boxed{Y_E = -1000 \text{ N}}$$

$$\theta = \tan^{-1} \left(\frac{2.5}{6} \right)$$

$$\theta = 22.29^\circ$$

In fig (II)

Member AB

$$F_A = F_B \quad \text{--- (IV)}$$

member CD

$$F_C = F_D \quad \text{--- (V)}$$

member BF

$$\sum M_F = F_D \times 5 \cos \theta + F_B \cos \theta \times 1.5 \quad \text{--- (VI)}$$

$$F_D = -F_B \times 1.5$$

$$\sum Y = F_B \sin \theta - F_D \sin \theta + Y_F = 0 \quad \text{--- (VII)}$$

$$F_B - F_D = 2702.7$$

$$F_B - (-F_B \times 1.5) = 2702.7$$

$$F_B + F_B \times 1.5 = 2702.7$$

$$F_B = \frac{2702.7}{2.5} = 1081.08 \text{ N}$$

$$F_D = -1081.08 \times 1.5 = -1621.62$$

$$\sum X = -F_B \cos \theta - F_D \cos \theta + X_F = 0 \quad \text{--- (VIII)}$$

$$(F_B + F_D) \cos \theta = X_F$$

$$(-540.14) \times 0.92 = X_F$$

$$X_F = -500.14 \text{ N}$$

In eq (I)

$$X_F + X_E = -600$$

$$X_E = -99.86 \text{ N}$$

$$F_B = 1081.08 \text{ N}$$

$$F_A = 1081.08 \text{ N}$$

$$F_D = -1621.62 \text{ N}$$

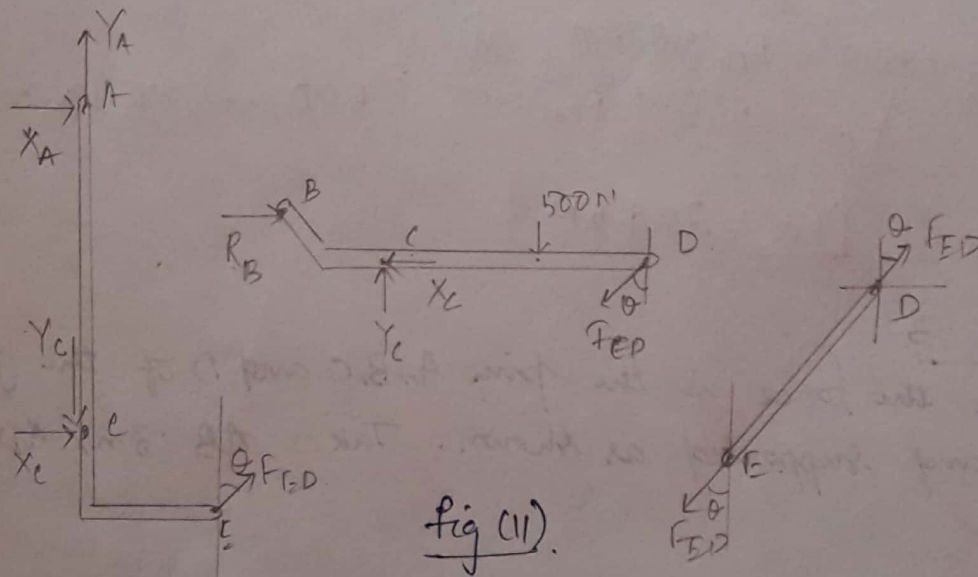
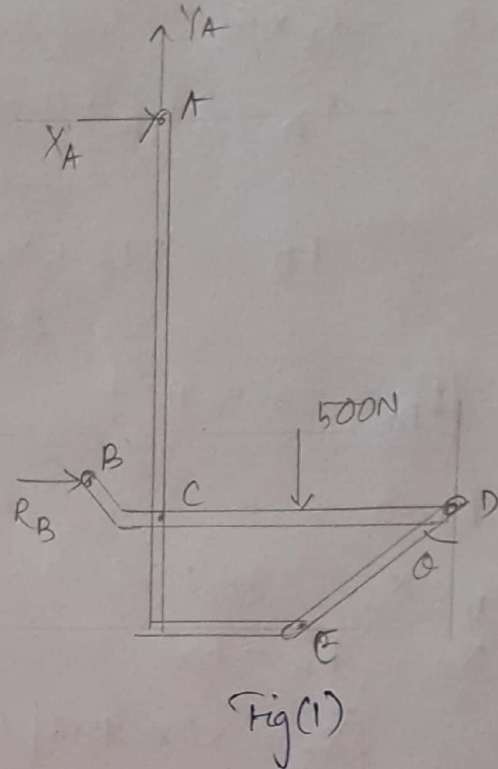
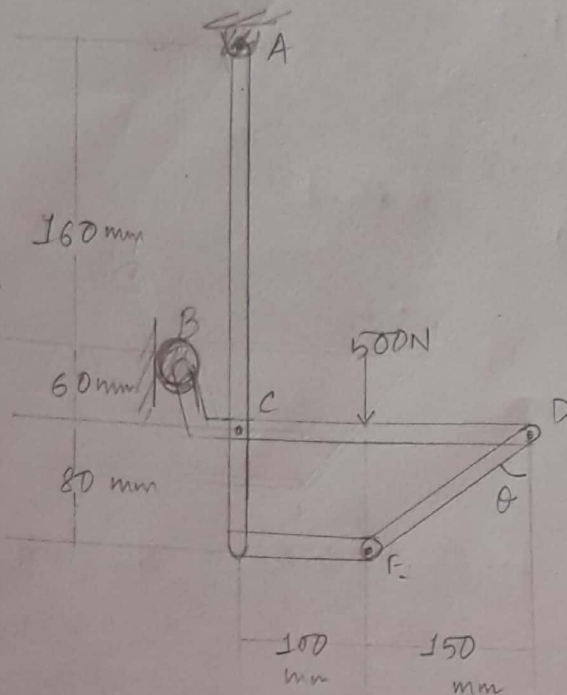
$$F_C = -1621.62 \text{ N}$$

$$X_F = -500.14 \text{ N}$$

$$X_E = -99.86 \text{ N}$$

Exercise - 2

Calculate the forces in the pins A, B, C, D and E of the frame loaded and supported as shown



$$\theta = \tan^{-1}\left(\frac{150}{80}\right) \Rightarrow \theta = 61.92^\circ$$

gm fig (11)

$$\sum F_x = 0 \quad X_A + R_B = 0 \quad \text{--- (i)}$$

$$\sum F_y = 0 \quad Y_A - 500 = 0 \quad \text{--- (ii)}$$

$$\sum M_A = 0 \quad R_B \times 160 = 500 \times 100 \quad \text{--- (iii)}$$

$$R_B = \frac{500 \times 100}{160} = 312.5$$

$$Y_A = 500 \text{ N}$$

$$R_B = 312.5 \text{ N}$$

$$X_A = -312.5 \text{ N}$$

In fig(11)

Member BD

$$\sum M_c = 0 \quad -R_B \times 60 - 500 \times 100 - F_{DE} \cos \theta \times 250 = 0 \quad \text{--- (W)}$$

$$-F_D = \frac{R_B \times 60 + 500 \times 100}{\cos \theta \times 250}$$

$$-F_D = \frac{312.5 \times 60 + 50000}{0.47 \times 250} = 585.1$$

$$\boxed{F_{DE} = -585.1 \text{ N}}$$

$$\sum F_x = 0$$

$$+R_B - X_C - F_{DE} \sin \theta = 0 \quad \text{--- (V)}$$

$$X_C = R_B - F_{DE} \sin \theta = 312.5 + 585.1 \times 0.88 = 828.72$$

$$\boxed{X_C = 828.72 \text{ N}}$$

$$\sum F_y = 0$$

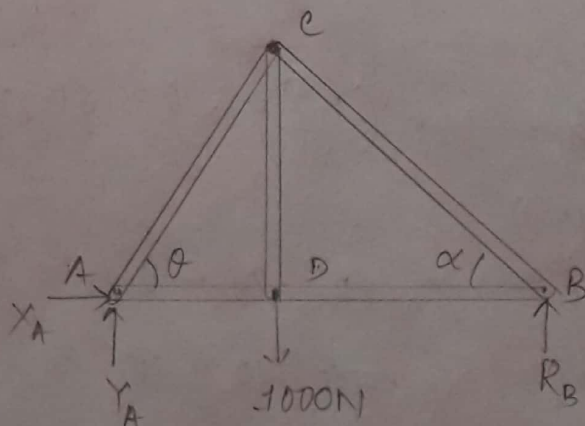
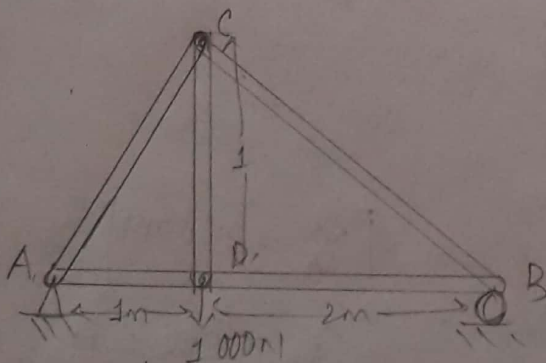
$$Y_C - 500 - F_{DE} \cos \theta = 0 \quad \text{--- (U)}$$

$$Y_C = 500 + F_{DE} \cos \theta = 500 - 585.1 \times 0.47 = 224.6$$

$$\boxed{Y_C = 224.6 \text{ N}}$$

Exercise - 3

Calculate the force in the pins A, B, C and D of the frame loaded and supported as shown. Take $AB = 3\text{m}$, $AD = 1\text{m}$, $CD = 1$.



Fig(1)

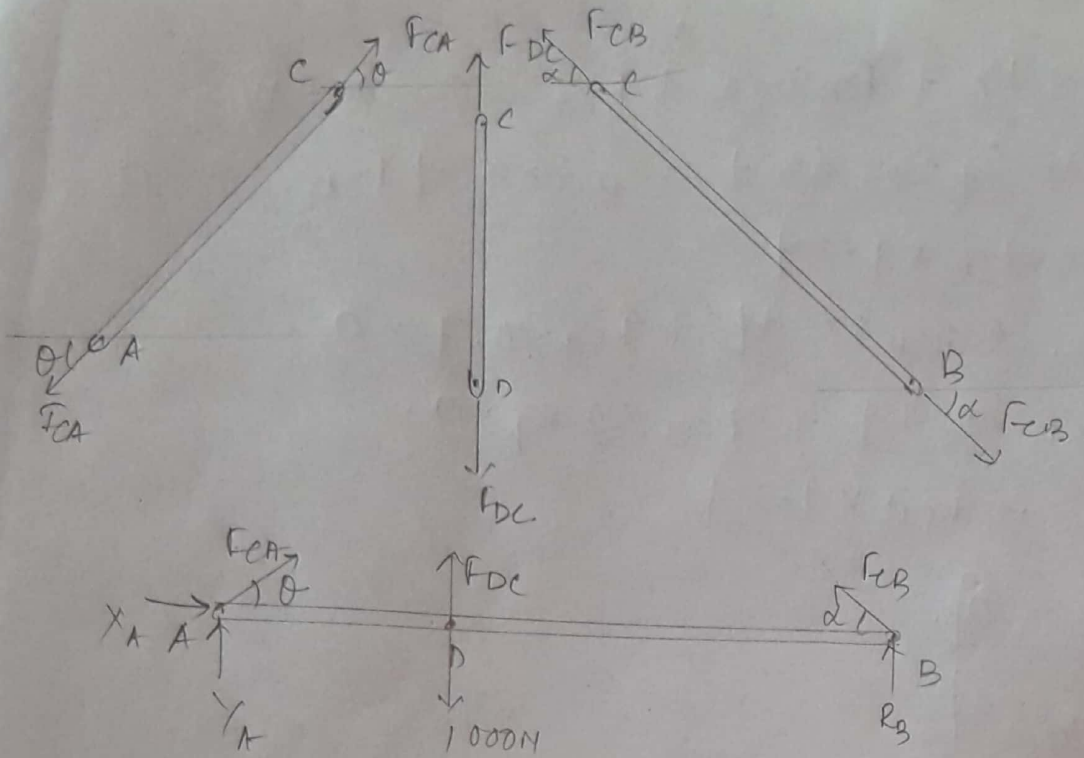


Fig (i)

In fig (i)

$$\theta = \tan^{-1}(1/1) = 45^\circ$$

$$\alpha = \tan^{-1}(1/2) = 26.56^\circ$$

$$\sum F_x = 0 \quad | \quad X_A = 0 \quad \text{--- (i)}$$

$$\sum M_A = 0 \quad -1000 \times 1 + R_B \times 3 = 0 \quad \text{--- (ii)}$$

$$| R_B = 333.33 \text{ N} |$$

$$R_B = 1000/3 = 333.33 \text{ N}$$

$$\sum F_y = 0 \quad Y_A - 1000 + R_B = 0 \quad \text{--- (iii)}$$

$$Y_A = 1000 - R_B = 1000 - 333.33$$

$$| Y_A = 666.67 \text{ N} |$$

In fig (ii)

$$| F_{DC} = 1000 | \quad \text{--- (iv)}$$

$$\sum F_x = 0 \quad X_A + F_{CA} \cos \theta - F_{CB} \cos \alpha = 0 \quad \text{--- (v)}$$

$$F_{CA} \times 0.7 = F_{CB} \times 0.89$$

$$| F_{CA} = 1.27 F_{CB} |$$

$$\sum M = 0$$

$$F_{DC} \times 1 - 1000 + F_{CB} \sin \alpha \times 3 + R_B \times 3 = 0 \quad \text{--- (vi)}$$

$$| F_{DC} = -F_{CB} \times 1.34 |$$

$$\sum F_y = 0$$

$$F_{DC} + Y_A - 1000 + R_B + F_{CB} \sin \alpha + F_{CA} \sin \theta = 0 \quad \text{--- (iv)}$$

$$666/666 - 1000 + 333/33 + F_{CB} \sin \alpha + F_{CA} \sin \theta = 0$$

$$F_{CA} = F_{CB} \times 1.34$$

$$F_{DC} + F_{CB} \times 0.44 + F_{CA} \times 0.7 = 0$$

$$F_{DC} + F_{CB} \times 0.44 + F_{CB} \times 0.89 = 0.$$

$$F_{DC} = -F_{CB} \times 1.34$$

$$\text{So, } \boxed{F_{DC} = 0}.$$

$$\boxed{F_{CB} = 0}$$

$$\boxed{F_{CA} = 0}$$

Exercise - 4

Calculate the forces in the pins A, B, C and D of the frame loaded and supported as shown. Assume $AD = 1.2\text{m}$, $DB = 1.2\text{m}$, $CD = 1\text{m}$, $DE = 1\text{m}$

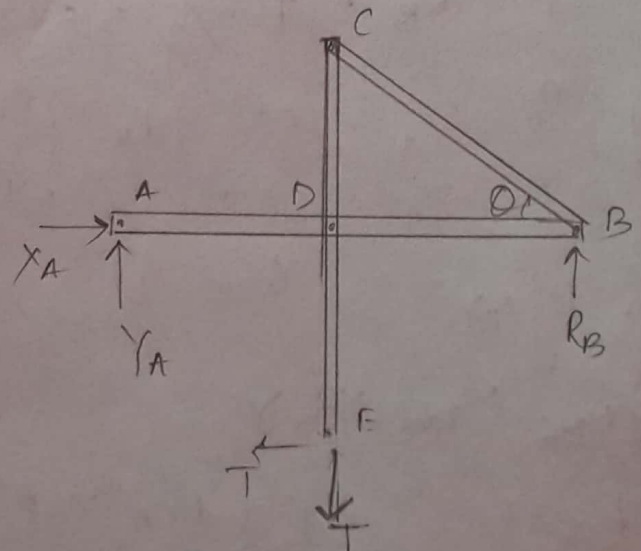
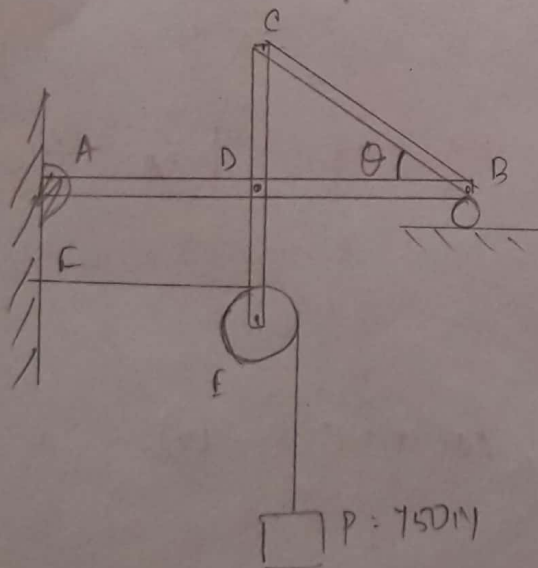


Fig. 1)

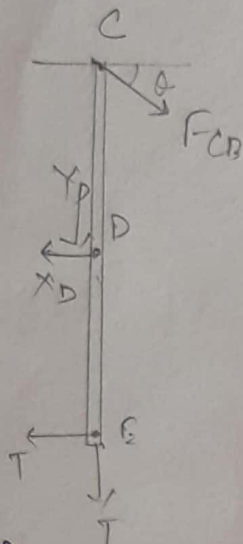
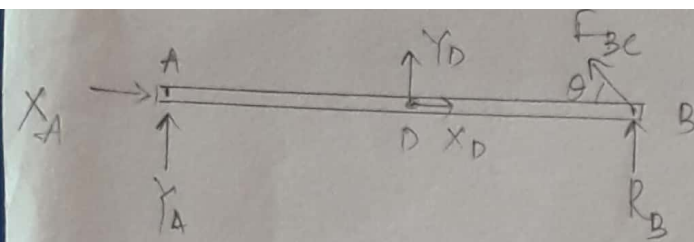
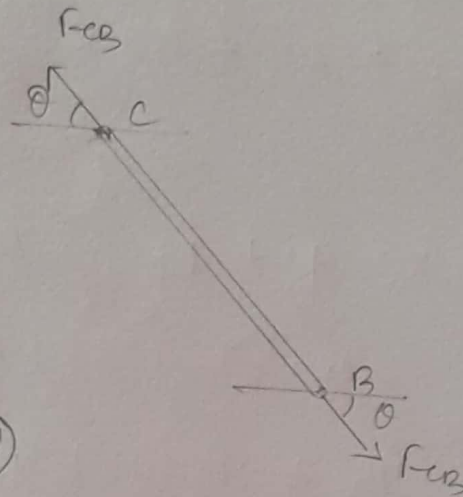


Fig (11)



In fig (11)

$$\theta = \tan^{-1}(1/1.2) \quad \theta = 39.69^\circ$$

$$T = 750 \text{ N}$$

$$\sum F_x = 0$$

$$X_A - T = 0 \quad \text{--- (i)}$$

$$X_A = 750 \text{ N}$$

$$\sum M_A = 0$$

$$-T \times 1.2 - T \times 1 + R_B \times 2.4 = 0 \quad \text{--- (ii)}$$

$$R_B = \frac{T \times 2.2}{2.4} = 687.5 \text{ N}$$

$$R_B = 687.5 \text{ N}$$

$$\sum F_y = 0$$

$$Y_A - T + R_B = 0 \quad \text{--- (iii)}$$

$$Y_A = T - R_B = 750 - 687.5$$

$$Y_A = 62.5 \text{ N}$$

In fig - II

$$\sum M_D = 0$$
$$-Y_A \times 1.2 + R_B \times 1.2 + R_{BC} \sin \theta \times 1.2 = 0 \quad \text{--- (W)}$$
$$-62.5 \times 1.2 + 687.5 \times 1.2 + F_{BC} \times 0.76 = 0$$

$$F_{BC} = \frac{-750}{0.76} = -986.84 \text{ N}$$

$$\boxed{F_{BC} = -986.84 \text{ N}}$$

$$\sum F_x = 0$$

$$X_A + X_D - F_B \cos \theta = 0 \quad \text{--- (V)}$$

$$750 + X_D + 986.84 \times 0.76 = 0$$

$$X_D = -1509.38 \text{ N}$$

$$\boxed{X_D = -1509.38 \text{ N}}$$

$$\sum F_y = 0$$

$$Y_A + Y_D + R_B + R_{BC} \sin \theta = 0 \quad \text{--- (U)}$$

$$62.5 + Y_D + 687.5 - 630.22 = 0$$

$$Y_D = -119.77 \text{ N}$$

$$\boxed{Y_D = -119.77 \text{ N}}$$